Equations And Inequalities

Chapter Overview

1. Simultaneous equations

Solve linear and quadratic systems using elimination or substitution.

2. Equations and graphs

Interpret simultaneous solutions as points of intersection.

3. Solving inequalities

Describe solution sets and combine conditions.

4. Inequalities on graphs

Compare curves and shade regions satisfying several inequalities.

Solutions sets

Solution 3

The solutions to an equation or inequality form its **solution set**. We consider real values here.

One value

$2x + 1 = 5$

One value

$\{2\}$

Several values

$x^2 + 3x + 2 = 0$

Several values

$\{-2, -1\}$

Infinitely many values

$x > 3$

Infinitely many values

$\{x \in \mathbb{R} : x > 3\}$

No real values

$x^2 = -1$

No real values

$\varnothing$

Every real value

$x^2 + x = x(x + 1)$

Every real value

$\mathbb{R}$

This identity is true for every real value of x.

Solutions sets

Solution 4

A simultaneous solution must satisfy every equation at the same time. Each solution is a pair (x, y).

One solution

$x + y = 9, \quad x - y = 1$

One solution

$\{(5, 4)\}$

Two solutions

$x^2 + y^2 = 10, \quad x + y = 4$

Two solutions

$\{(3, 1), (1, 3)\}$

No solutions

$x + y = 1, \quad x + y = 3$

No solutions

$\varnothing$

The two equations cannot both hold.

Infinitely many solutions

$x + y = 1, \quad 2x + 2y = 2$

Infinitely many solutions

$\{(x, 1 - x) : x \in \mathbb{R}\}$

For example: $(0, 1), (1, 0), (2, -1), (0.5, 0.5)$.

1 :: Simultaneous Equations

Solution 5

1. Elimination Solve $3x + y = 8$ and $2x - 3y = 9$.

1. Elimination

$$9x + 3y = 24$$
$$2x - 3y = 9$$
$$11x = 33 \Rightarrow x = 3$$
$$y = 8 - 3(3) = -1$$

2. Substitution Solve $x + 2y = 3$ and $x^2 + 3xy = 10$.

2. Substitution

$$x = 3 - 2y$$
$$(3 - 2y)^2 + 3y(3 - 2y) = 10$$
$$2y^2 + 3y + 1 = 0$$
$$(2y + 1)(y + 1) = 0$$

The pairs are $(4, -1/2)$ and $(5, -1)$.

Elimination is convenient for linear equations. For a linear–quadratic system, rearrange the linear equation and substitute into the quadratic. Check each pair in both original equations.

Test Your Understanding

Solution 6

Solve simultaneously

$3x^2 + y^2 = 21, \qquad y = x + 1$

Solve simultaneously

$$3x^2 + (x + 1)^2 = 21$$
$$4x^2 + 2x - 20 = 0$$
$$(2x + 5)(x - 2) = 0$$
$$(x, y) = (-\tfrac{5}{2}, -\tfrac{3}{2}) \quad \text{or} \quad (2, 3)$$

Exercise 3A/B – Y1 Pages 40, 41 Solution 7

4 Solve $x + y = 3$ and $x^2 - 3y = 1$.

4

$$y = 3 - x$$
$$x^2 - 3(3 - x) = 1$$
$$x^2 + 3x - 10 = 0$$
$$(x + 5)(x - 2) = 0$$
$$(x, y) = (-5, 8) \quad \text{or} \quad (2, 1)$$

Exercise 3A/B – Y1 Pages 40, 41 Solution 8

5. The equations are $y = 2 - 4x$ and $3x^2 + xy + 11 = 0$.

a Eliminate y and show that $x^2 - 2x - 11 = 0$.

a

$$3x^2 + x(2 - 4x) + 11 = 0 \Rightarrow x^2 - 2x - 11 = 0$$

b Solve the equations, giving exact answers involving $\sqrt{3}$.

b

$$x = \frac{2 \pm \sqrt{48}}{2} = 1 \pm 2\sqrt{3}$$
$$(x, y) = (1 + 2\sqrt{3}, -2 - 8\sqrt{3})$$
$$\text{or } (1 - 2\sqrt{3}, -2 + 8\sqrt{3}).$$

Exercise 3A/B – Y1 Pages 40, 41 Solution 9

6. One solution of $y = kx - 5$ and $4x^2 - xy = 6$ is $(1, p)$.

a Find k and p.

a

$4 - p = 6 \Rightarrow p = -2, \qquad p = k - 5 \Rightarrow k = 3$

b Find the other solution.

b

$4x^2 - x(3x - 5) = 6$

$x^2 + 5x - 6 = 0$

$(x - 1)(x + 6) = 0$

The second pair is $(-6, -23)$.

Extension Solution 10

1. STEP 2010 Q1

Given the identity

$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$

1(a) Find a, b, c, d.

1(a)

Compare coefficients after expansion:

$a = 1, \quad b = 1, \quad c = -2, \quad d = -4$

1(b) Solve

$5x^2 + 2y^2 - 6xy + 4x - 4y = 9$

$6x^2 + 3y^2 - 8xy + 8x - 8y = 14.$

1(b)

Write $U = x - y + 2$, $V = y - 2x$. Then $U^2 + V^2 = 13$ and $2U^2 + V^2 = 22$. Thus $U = \pm 3, V = \pm 2$.

$(x, y) \in \{(-3, -4), (1, 0), (3, 8), (7, 12)\}$

2. MAT 2012 1G For which k do positive real x, y satisfy $2x + ky = 4$, $x + y = k$?

A: all values; B: none; C: only 2; D: only $k > -2$.

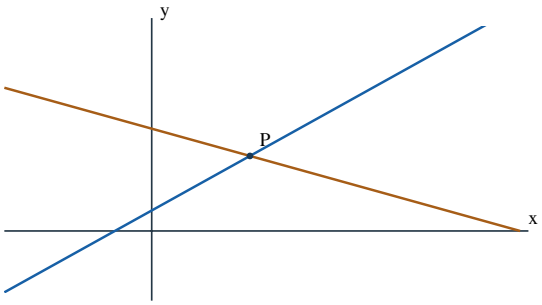
2. MAT 2012 1G

Substitute $x = k - y$: $(k - 2)(y + 2) = 0$. Since $y > 0$, we need $k = 2$. Positive pairs with $x + y = 2$ exist. Answer C.

Simultaneous Equations and Graphs

Solution 11

The graph of an equation contains all points (x, y) satisfying it. For example, $(3, 7)$ lies on $y = 2x + 1$.



Two simultaneous equations

$y = 2x + 1, \qquad x + y = 5$

Which point satisfies both equations?

Two simultaneous equations

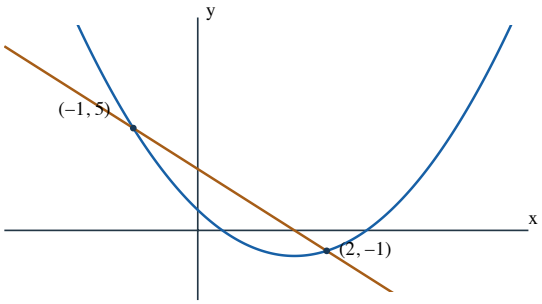
The point of intersection lies on both graphs.

$3x + 1 = 5 \Rightarrow (x, y) = (\frac{4}{3}, \frac{11}{3})$

Example

Solution 12

On the same axes, draw $y = 3 - 2x$ and $y = x^2 - 3x + 1$.



a. Intersections Use the graph to solve the simultaneous equations.

a. Intersections

$(x, y) = (-1, 5) \quad \text{or} \quad (2, -1)$

b. Algebraic check How can we prove that there are two intersections?

b. Algebraic check

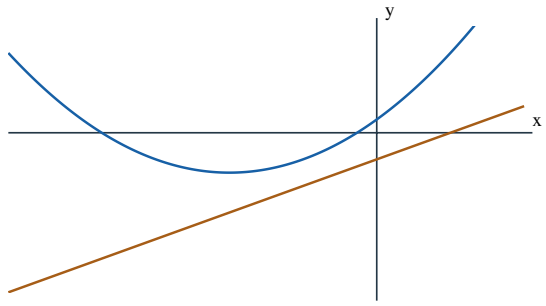
$x^2 - 3x + 1 = 3 - 2x \Rightarrow x^2 - x - 2 = 0$

The discriminant is $1 + 8 = 9 > 0$, so there are two distinct real solutions.

Another Example

Solution 13

Draw $y = 2x - 2$ and $y = x^2 + 4x + 1$ on the same axes.



Prove the graphs never meet Solve simultaneously and consider the discriminant.

Prove the graphs never meet

$x^2 + 4x + 1 = 2x - 2$

$x^2 + 2x + 3 = 0$

$b^2 - 4ac = 4 - 12 = -8 < 0$

There are no real solutions and no intersections.

Final Example

Solution 14

The line $y = 2x + 1$ meets $kx^2 + 2y + (k - 2) = 0$ at exactly one point. The constant k is positive.

a Find k.

a

$kx^2 + 2(2x + 1) + k - 2 = 0 \Rightarrow kx^2 + 4x + k = 0$

Exactly one real root requires

$16 - 4k^2 = 0 \Rightarrow k = \pm 2.$

Since $k > 0$, $k = 2$.

b Find the point of intersection.

b

$2x^2 + 4x + 2 = 0 \Rightarrow (x + 1)^2 = 0$

$x = -1, \qquad y = -1 \Rightarrow (-1, -1)$

Exercise 3C – Y1 Page 45 — Solution 15

8. Given $2x - y = 1$, $x^2 + 4ky + 5k = 0$, where $k \neq 0$.

a Show that $x^2 + 8kx + k = 0$.

a

$y = 2x - 1 \Rightarrow x^2 + 4k(2x - 1) + 5k = 0$

b Given equal roots, find k.

b

$64k^2 - 4k = 0 \Rightarrow 4k(16k - 1) = 0$

$k = \frac{1}{16} \quad (k \neq 0)$

c Find the simultaneous solution.

c

$16x^2 + 8x + 1 = (4x + 1)^2 = 0$

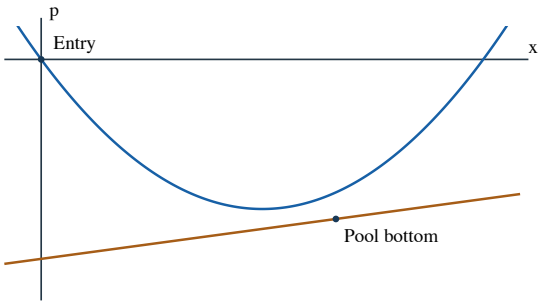
$(x, y) = (-\frac{1}{4}, -\frac{3}{2})$

Exercise 3C – Y1 Page 45 — Solution 16

9. A swimmer dives into a pool. Her vertical position p metres is modelled in terms of horizontal distance x metres from entry by

$p = \frac{1}{2}x^2 - 3x$.

The pool bottom is modelled by $p = 0.3x - 6$. Determine whether the swimmer touches the bottom.



9

$0.5x^2 - 3x = 0.3x - 6$

$0.5x^2 - 3.3x + 6 = 0$

$b^2 - 4ac = 10.89 - 12 = -1.11 < 0$

The graphs never intersect. The model predicts that she does not touch the pool bottom.

Set Builder Notation

Solution 17

A **set** is a collection of values. Order does not matter and there are no duplicates. Use braces, e.g. $A = \{1, 4, 6, 7\}$.

$A \cap B$: intersection, elements in both sets.

$A \cup B$: union, elements in either set or both.

$\varnothing$: the empty set.

$\mathbb{N}$: positive integers (our convention here).

$\mathbb{Z}$: all integers.

$\mathbb{R}$: all real numbers.

$x \in A$: x is a member of A.

1

$\{1, 2, 3\} \cap \{3, 4, 5\}$

1

$\{3\}$

2

$\{1, 2, 3\} \cup \{3, 4, 5\}$

2

$\{1, 2, 3, 4, 5\}$

3

$\{1, 2\} \cap \{3, 4\}$

3

$\varnothing$

Set Builder Notation

Solution 18

Set builder notation describes values without listing them:

{expression : condition}

A colon or vertical bar means “such that”. What do these sets contain?

a

$\{2x : x \in \mathbb{Z}\}$

a

All even integers:

$\{\ldots, -4, -2, 0, 2, 4, \ldots\}$

b

$\{2^x : x \in \mathbb{N}\}$

b

Positive-integer powers of 2:

$\{2, 4, 8, 16, 32, \ldots\}$

c

$\{xy : x, y \text{ are prime}\}$

c

Products of two primes:

$\{4, 6, 9, 10, 14, 15, \ldots\}$

Set builder notation is useful for describing solution sets.

Set Builder Notation

Solution 19

Use set builder notation to specify each set. Unless stated otherwise, x is real.

1 All odd integers.

1

$\{2n+1 : n \in \mathbb{Z}\}$

2 All real numbers greater than 5.

2

$\{x \in \mathbb{R} : x > 5\}$

3 All real numbers less than 5 or greater than 7.

3

$\{x : x < 5\} \cup \{x : x > 7\}$

4 All real numbers between 5 and 7 inclusive.

4

$\{x : 5 \leq x \leq 7\}$

Equivalently, $\{x : x \geq 5\} \cap \{x : x \leq 7\}$.

Recap of linear inequalities

Solution 20

1

$2x + 1 > 5$

1

$2x > 4 \Rightarrow \{x : x > 2\}$

2

$3(x-5) \geq 5 - 2(x-8)$

2

$5x \geq 36 \Rightarrow \{x : x \geq 7.2\}$

3

$-x \geq 2$

3

$\{x : x \leq -2\}$

Multiplying or dividing both sides by a negative number reverses the inequality sign.

Combining inequalities If $x < 3$ and $2 \leq x < 4$, what values satisfy both?

Combining inequalities

$2 \leq x < 3$

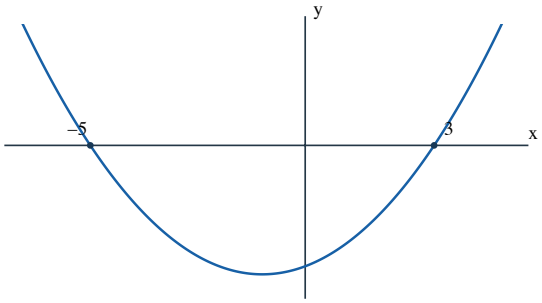
Take the intersection of the two solution sets.

RECAP :: Solving Quadratic Inequalities

Solution 21

Solve $x^2 + 2x - 15 > 0$.

First put zero on one side, then factorise: $(x + 5)(x - 3) > 0$.



Solution

The curve is above the x-axis outside its two roots.

$x < -5 \quad \text{or} \quad x > 3$

In set notation: $\{x : x < -5\} \cup \{x : x > 3\}$.

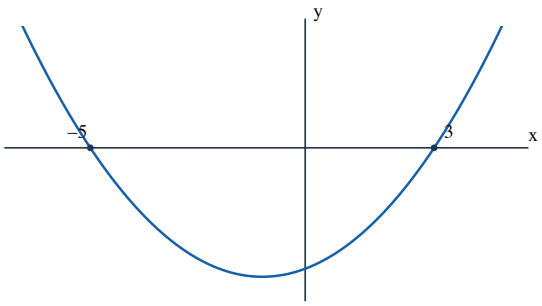
Strict inequality means the roots are excluded.

Solving Quadratic Inequalities

Solution 22

Solve $x^2 + 2x - 15 \leq 0$.

Factorise: $(x + 5)(x - 3) \leq 0$.



Solution

The curve is on or below the x-axis between the roots.

$\{x : -5 \leq x \leq 3\}$

The non-strict inequality includes both endpoints.

Further Examples

Solution 23

1

$x^2 + 5x + 4 \geq 0$

1

$(x+4)(x+1) \geq 0$

The upward parabola is nonnegative outside the roots:

$x \leq -4 \quad \text{or} \quad x \geq -1$

2

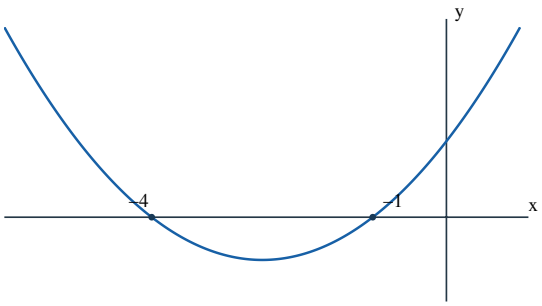
$x^2 - 9 < 0$

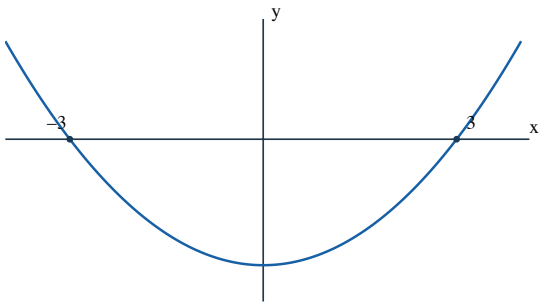
2

$(x+3)(x-3) < 0$

The upward parabola is negative between the roots:

$-3 < x < 3$





Sketch the graph to justify which intervals work. A calculator inequality solver can check your result; choose degree 2.

Test Your Understanding

Solution 24

1. Find the set of values of x satisfying each condition.

1(a)

$3(x-2) < 8 - 2x$

1(a)

$5x < 14 \Rightarrow x < \frac{14}{5}$

1(b)

$(2x-7)(1+x) < 0$

1(b)

$-1 < x < \frac{7}{2}$

1(c) Both inequalities above.

1(c)

$-1 < x < \frac{14}{5}$

2. The quadratic equation $2qx^2 + qx - 1 = 0$, with $q \neq 0$, has no real roots.

2(a) Show that $q^2 + 8q < 0$.

2(a)

$b^2 - 4ac = q^2 + 8q < 0$

2(b) Find the possible values of q.

2(b)

$q(q+8) < 0 \Rightarrow -8 < q < 0$

We solve an inequality in q to identify equations that have no real solutions in x. These are different variables and different solution sets.

Deal with inequalities with a division by x

Solution 25

Find the values satisfying $6/x > 2$, where $x \neq 0$.

Do not simply multiply by x: its unknown sign determines whether the inequality reverses. Instead multiply by $x^2 > 0$.

Working Multiply both sides by x^2, then solve.

Working

$$6x > 2x^2$$
$$x(x-3) < 0$$
$$0 < x < 3$$

Alternatively compare the graphs $y = 6/x$ and $y = 2$. Always keep the original restriction $x \neq 0$.

Exercise 3D/3E – Y1 Page 47-48, 50-51

Solution 26

3(g) Use set notation:

$4(3x+7) \leq 20$ or $2(3x-5) \geq \dfrac{7-6x}{2}$

3(g)

$12x \leq -8$ or $18x \geq 27$

$\{x : x \leq -\frac{2}{3}\} \cup \{x : x \geq \frac{3}{2}\}$

6 Find the values satisfying $x^2 - 5x - 14 > 0$.

6

$(x-7)(x+2) > 0$

An upward parabola is positive outside the roots:

$\{x : x < -2\} \cup \{x : x > 7\}$

8 Given $x \neq 3$, solve $5/(x-3) < 2$.

8

Multiply by $(x-3)^2 > 0$:

$$5(x-3) < 2(x-3)^2$$

$$2x^2 - 17x + 33 > 0$$

$$(2x-11)(x-3) > 0$$

$$\{x : x < 3\} \cup \{x : x > \tfrac{11}{2}\}$$

Both endpoints are excluded. At x=3 the original fraction is undefined; at x=11/2 the two sides are equal.

9 The equation $kx^2 - 2kx + 3 = 0$ has no real roots. Prove that $0 \leq k < 3$.

9

First consider $k = 0$: the equation becomes $3 = 0$, so it has no solutions.

For $k \neq 0$, it is quadratic and requires

$$(-2k)^2 - 4(k)(3) < 0$$

$$4k(k-3) < 0$$

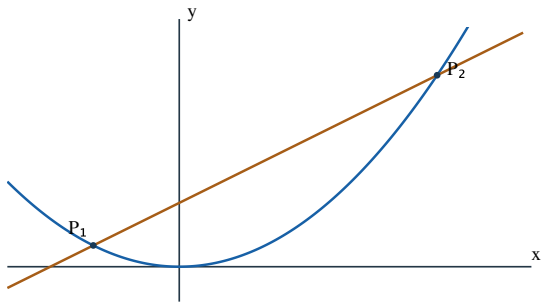
$$0 < k < 3.$$

Combining the cases gives $0 \leq k < 3$.

Inequalities on Graphs

Solution 29

Compare $L_1 : y = 12 + 4x$ and $L_2 : y = x^2$.



a Find their intersections.

a

$x^2 - 4x - 12 = (x - 6)(x + 2) = 0$

$P_1 = (-2, 4), \quad P_2 = (6, 36)$

b Hence solve $12 + 4x > x^2$.

b

The line lies above the curve between the intersections:

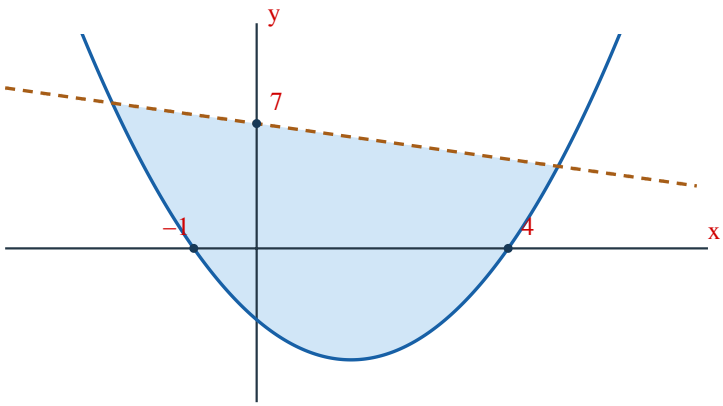
$-2 < x < 6$

Inequality Regions

Solution 30

Shade the region satisfying $2y + x < 14$ and $y \geq x^2 - 3x - 4$.

Sketch and shade



1. Draw the boundaries

$y = 7 - x/2$: dashed, since the inequality is strict. Its intercepts are $(0, 7)$ and $(14, 0)$.

$y = x^2 - 3x - 4$: solid, since equality is allowed.

Shaded region

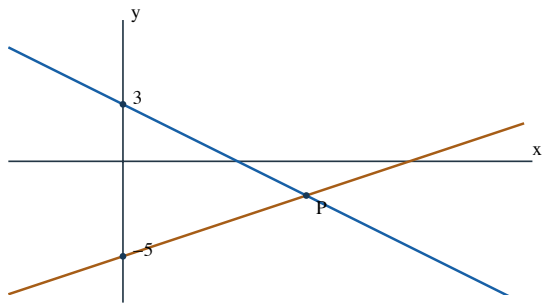
2. Choose the common region

Shade above or on the parabola and strictly below the straight line.

The shaded region exists between the intersections, with the line itself excluded.

Exercise 3F/3G – Y1 Page 53, 55 — Solution 31

1. The lines are $L_1 : 2y + 3x = 6$ and $L_2 : x - y = 5$.



a Find their point of intersection P.

a

$$3x + 2y = 6$$
$$2x - 2y = 10$$
$$5x = 16$$
$$P = (\tfrac{16}{5}, -\tfrac{9}{5})$$

b Find where L_1 is above L_2, that is, solve $(6 - 3x)/2 > x - 5$.

b

$$x < \tfrac{16}{5}$$

Exercise 3F/3G – Y1 Page 53, 55 — Solution 32

3. Find where $y = f(x)$ is strictly below $y = g(x)$.

a

$f(x) = 3x^2 - 2x - 1, \quad g(x) = x + 5$

a

$$3x^2 - 3x - 6 = 3(x - 2)(x + 1) < 0$$
$$-1 < x < 2$$

b

$f(x) = 2x^2 - 4x + 1, \quad g(x) = 3x - 2$

b

$$2x^2 - 7x + 3 = (2x - 1)(x - 3) < 0$$
$$\tfrac{1}{2} < x < 3$$

c

$f(x) = 5x - 2x^2 - 4, \quad g(x) = -2x - 1$

c

$$-2x^2 + 7x - 3 < 0 \Rightarrow (2x - 1)(x - 3) > 0$$
$$x < \tfrac{1}{2} \quad \text{or} \quad x > 3$$